

RAMCO INSTITUTE OF TECHNOLOGY
Department of Mechanical Engineering
Academic Year 2019 – 2020 (Odd Semester)

Innovative practice(s) description

Course code and title: OME551 Energy Conservation and Management

Name of the Instructor(s): Mr.R.Arun Kumar, AP//Mechanical

Date & Time: 16.07.2019 & 9:50AM to 10:15AM

Field visit and energy auditing assignment

As energy conservation becomes the need of the hour, students were taught about its importance and ways to achieve it. The two ways to achieve energy conservation are:

- Generation of electricity from green energy sources (renewable energy sources)
- Effective and optimized utilization of available energy resources

In order to make the students to get clear idea about above mentioned two ways of energy conservation, they underwent a field visit and were assigned an individual assignment on energy auditing.

Students underwent a **field visit to have a look on 30kVA roof top Solar PV plant** installed in western wing of institution premises. There they were explained about type of PV panel used, utilization procedure and benefits of installed plant towards energy conservation. Then students were given an **individual assignment on energy auditing** in which they have to audit the electrical utilities in their respective home or hostel block. Students have submitted an individual report highlighting the energy consumption pattern of available electrical utilities in the premises they have audited and suggested possible ways of conserving the energy.

OUTCOME:

Through this activity the students are made to understand the possible ways of conserving energy as mentioned above with an practical exposure which have given them an insight knowledge.

This activity helps in attaining the following CO – PO mapping:

Course Outcome / Programme Outcome	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO11	PO12
CO1 – Student will be able to explain energy auditing methodology	2	1	1	1	2	1	2	2	2	1



ENERGY AUDIT REPORT

NAME OF THE AUDITOR : Miss. V. SINDHUJA

PLACE OF AUDIT : Home

ADDRESS : 266/2, West Street,
Theetham Patti,
Kovilpatti (TN)

DATE OF AUDIT : 22.07.19

TYPE OF AUDIT : SIMPLE AUDIT (WALK THROUGH AUDIT)

AUDIT STATEMENT

MAJOR UTILITIES IN HOME.

S.no	utility	Capacity	Quantity/Nos	Power Consumption
1.	LCD Television	-	1	190 W-hr
2.	Air Conditioner	1 tonne	1	1000 W-hr
3.	Washing Machine	-	1	500 W-hr
4.	Refrigerator	-	1	120 W-hr
5.	Electric Iron	-	1	150 W-hr

Other Utilities in Home:

S.no	utility	Capacity	Quantity/Nos	Power Consumption
1.	Bulb	-	15	60 W-hr
2.	Tube light	-	6	40 W-hr
3.	Night lamps	-	3	1 W-hr
4.	CFLs	-	4	15 W-hr

S-NO.	Utility	Running time / Day	Power Consumption per day
1.	Refrigerator	5 hr	0.600 kW-hr
2.	LCD Television	4 hr	0.76 kW-hr
3.	Air conditioner	6 hr	6 kW-hr
4.	Tube light	5 hr x 6	1.20 kW-hr
5.	CFL's	7 hr x 4	4.2 kW-hr
			12.76 units.

Scope of power Conservation:

* Air Conditioner can be replaced with Inverter AC which consumes only 50% of Conventional AC's.

* use halogen Incandescent bulbs / Compact Fluorescent lights and light emitting diodes bulbs which requires only 25-80% less electricity and last three to 25 times longer than traditional bulbs.

* A smart thermostat can be set to automatically turn off or reduce heating and cooling during the times when we are asleep or away. It can save upto 10% of electricity.

* use LED display television which consumes 90W per hour.

* use zero watts bulb instead of any bulbs which can conserve upto 10% of electricity.

Note: The above replacement consumers only 10 units per day (i.e) nearly 60% of energy can be saved.